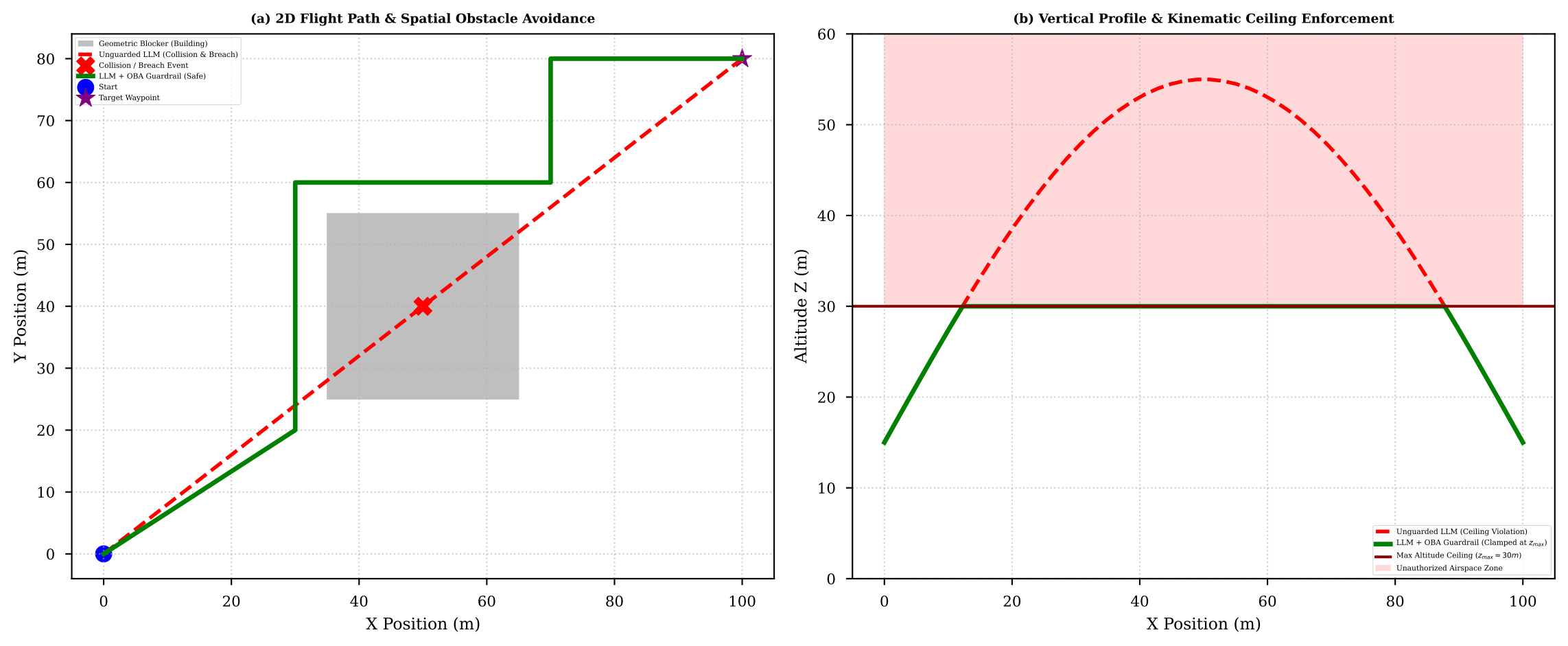
(a) 2D Flight Path & Spatial Obstacle Avoidance
Geometric Blocker (Building)
Unguarded LLM (Collision & Breach)
Collision / Breach Event
LLM + OBA Guardrail (Safe)
Start
Target Waypoint
Y Position (m)
X Position (m)
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100
(b) Vertical Profile & Kinematic Ceiling Enforcement
Altitude Z (m)
X Position (m)
0
10
20
30
40
50
60
0
20
40
60
80
100
Unguarded LLM (Ceiling Violation)
LLM + OBA Guardrail (Clamped at z_{max})
Max Altitude Ceiling ($z_{max} = 30m$)
Unauthorized Airspace Zone